Measurement 259 (2026) 119644

Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement

In situ stress monitoring and calibration of fiber Bragg Gratings embedded inside aluminum samples at high temperatures

Constantin Bauer [a,*], Robert Kleinhans [b], Maximilian Erber [a], Manuel Pintore [b], Christoph Hartmann [a], Wolfram Volk [a,b]

[a] Chair of Metal Forming and Casting, Technical University of Munich, Walther-Meissner-Str. 4, Garching, 85748, Germany
[b] IGCV Fraunhofer, Lichtenbergstraße 15, Garching, 85748, Germany

ARTICLE INFO

Keywords:
Aluminum casting
Fiber Bragg Grating
Star mold geometry
High-temperature tensile testing
Hot tearing

ABSTRACT

Hot tears are one of many defects in castings and weldings. Especially in weldings, this crack-like defect leads to failure or rejection in production. Nevertheless, castings are also being affected by this phenomenon. Several attempts exist to understand and describe hot tearing, but only a few methods exist to analyze the effect in situ. Fiber Bragg Gratings (*FBGs*) have been used in castings to measure strain and are now used to inspect hot tearing in situ. With a unique test bench, built of a mirror furnace capable of heating specimens and a tensile testing machine that can pull the prepared specimen simultaneously, it is now possible to calibrate fiber Bragg Gratings inside aluminum samples and analyze the formation of cracks with the optical method of FBGs. This experiment allows a defined application and variation of stress on the aluminum sample made of different alloys, and conducts tests at varying temperatures up to the solidus temperature. To get a valid measurement, it is necessary to calibrate the fibers precisely. This task is not trivial since the Bragg Grating is sensitive to changes in temperature and strain. In this paper, we controlled the furnace to sustain a defined temperature and applied increasing stress steps to the sample. In this way, we can differentiate between the changes caused by the increasing stress and the changes from the rise in temperature. This calibration approach makes it possible to more easily detect hot tearing around the line of solidus for hypo-eutectic and hyper-eutectic alloys with a wide solidification range.

1. Introduction

Metal casting is a process that consists of different steps. The two main processing steps are pouring and solidifying the metal. Both process steps have great optimization potential; wrong handling during these steps can cause several casting defects. The following research focuses on the second step and examines the effects of solidification, conducting measurements during this process, at higher temperature ranges. Hereby, an optical fiber sensing technology performs enhanced measurements inside liquid aluminum to observe the effects of the solidification in situ. Various works on the subject exist, and the successes that can be achieved with fiber optic Bragg Grating (*FBG*) measurements inside liquid aluminum prove the technology's applicability inside castings [1,2]. Recent research shows that calibrating FBGs inside castings is possible [3]. It is also possible to use this technology to monitor strain inside castings at more than one position [4] or repeatedly remelt metal samples [5]. Recent papers focus on measuring the stress occurring from the solidification. This paper uses a tensile testing machine to apply stress actively to the samples at high temperatures. Experiments are performed inside a lab environment to get respective results. The mirror furnace is also used for high-temperature treatment [6]. Similar approaches with high temperature testing are conducted for composite structures [7] or the tensile testing for cylindrical steel samples [8]. Other research uses a similar setup, but it is not possible to create hot spots with this furnace [9]. One disadvantage is that the use of a laser Extensometer is not possible when working with a mirror furnace because of the light intensity. Because the investigation focuses on the mechanism of hot cracking, hot spots are needed to force this phenomenon to happen [10]. Fig. 1 shows a phase diagram with the temperature on the ordinate and the mass percentage of silicon on the abscissa. On the right, the legend indicates in which areas it is more likely to create hot tears. Typically, hot tears are defined to occur between the line of solidus and the line of liquidus [11,12]. They occur in the mushy zone when the solidification shrinkage causes tensile stress to the growing dendrites [13], and different approaches exist for modeling the defect [14]. The chances of developing hot tears

* Corresponding author.
E-mail address: c.bauer@tum.de (C. Bauer).

https://doi.org/10.1016/j.measurement.2025.119644
Received 22 April 2025; Received in revised form 2 October 2025; Accepted 6 November 2025
Available online 11 November 2025
0263-2241/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

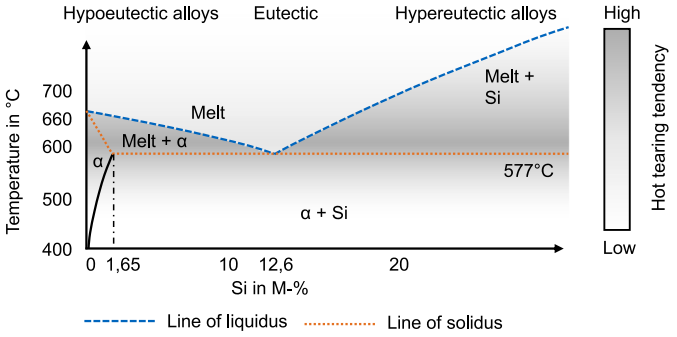


Fig. 1. Aluminum–silicon phase diagram highlighting the regions of hot tearing tendency occurring during solidification.

increase the closer the temperature is to the line of solidus. At this temperature interval, it is difficult to conduct stress measurements in situ, but one solution is offered by FBGs. How this sensor works is therefore explained in the next section.

1.1. Fiber optic sensing

Fiber optic sensing is used both for process monitoring as well as life cycle observations [15]. FBGs can be imagined as optical strain gauges. Therefore, they do not depend on electricity, which makes them robust against external magnetic fields or other harsh environmental influences. Another advantage is that fibers only need a single connector, instead of several cables [16]. Even when working with arrays of FBGs, only a single fiber is enough to transfer the information along the fiber. For high temperature measurements, glass fibers of the type SMF-28 are used, but fibers can have different base materials [17]. For castings, typically because of the high temperatures, and the applicability, fibers made of glass are used, but many other kinds of fibers exist with different advantages and disadvantages [18,19]. Fig. 2 shows the fundamental structure of a step-indexed single-mode fiber. For almost every fiber, it is true that they have a cylindrical-based shape. Which is why they can be easiest characterized in cylindrical coordinates. A step-indexed single-mode fiber is represented in Fig. 2, showing the profile of the numerical aperture (a). The radial coordinate is ρ, and n is the refractive index. The core radius a_K of the fiber at which it has a refractive index of n_K is represented. At the coating, the refractive index is described by n_M. The coating's purpose is to protect the fiber from getting any scratches, because the fiber consists of quartz glass (SiO_2). Fig. 2 (b) shows a three-dimensional rendering of the fiber. The core typically has a diameter of about 8 µm whereas the optical fiber itself has a diameter of about 125 µm, which is about twice as thick as a human's hair. The outer coating is made of an organic material and has only protective purposes, which is why it is removed when the fiber is placed inside liquid aluminum. In some cases, this coating cannot be removed, but it burns as soon as it comes into contact with liquid aluminum. For measuring tasks, the fiber is used to transmit light at the wavelength of a bandwidth of 1525 – 1570 nm. The broad input spectrum is generated by an optical device called an interrogator. The light travels along the axial direction as long as it meets the grating placed inside the core. At this point, the grating reflects a specific wavelength called λ_B which is calculated from the following Eq. (1). Whereas n_{eff} is the effective refractive index of the fiber and Λ_{FBG} is the grating periodicity.

$$\lambda_B = 2 n_{eff} \Lambda_{FBG} \tag{1}$$

The remaining spectrum is transmitted to the end of the fiber and has no relevance to the measurement. Other than the reflected spectrum that the interrogator detects. The reflected spectrum changes anytime the spacing of the grating Λ_{FBG} changes. A cause could be stress to the fiber, which leads to changes in the spacing of filter elements. However, stress can be driven by temperature changes affecting the fiber or by forces acting on the fiber. This causes a cross-sensitivity to temperature and strain and is a real challenge to the applicability of FBGs. [20–22]

There are different ways of placing the optical filter inside the fiber. For the experiments, we used standardized femto-second pulsed FBGs. They are easily available because they are produced by an automated manufacturing technology. Their key properties are that they can resist temperatures of up to 1000 °C and are immune to humidity.

Table 1
List of elements measured by optical emission spectroscopy of the two aluminum alloys investigated.

Al. alloy	OES result	Al in %	Si in %	Cu in %	Fe in %	Mn, Mg, Zn others in %
AlSi5 n=5	Mean	93.8	5.06	0.009	0.110	≤0.128
AlSi17 n=5	Mean	82.5	17.11	0.005	0.141	≤ 0.105

1.2. Preparation of the specimen

Because hot tearing is stronger for alloys with a broader solidification range, the two alloys Al-5wt.%Si (*AlSi5*) and Al-17wt.%Si (*AlSi17*) are chosen for deeper investigation. The following Table 1 gives the measured distribution of the alloy elements in weight percent inside the aluminum material. The values were measured by optical emission spectroscopy (*OES*) on the samples at different positions, and the mean value is then calculated over the sample size of n.

The lab samples for the calibration experiments are prepared in two different ways in order to position the glass fiber inside the aluminum. One way of preparing is by gravity casting the samples. Therefore, a casting system shown in Fig. 3 (a) is manufactured to create a cylindrical base geometry. The casting system is made of an inlet (1), a sprue (2), and a runner (3). The runner is also the ingate to the cylindrical base geometry from which the sample (6) is later machined. On top of this geometry sits the feeder (5). The glass fiber (7) is positioned before casting inside the empty mold along the symmetric center of the cylindrical part. Just as highlighted in Fig. 3 (a) by number (7) and the yellow line. The second way, which was tested to manufacture samples, is to machine the samples first, and instead of casting the aluminum around the fiber, little boreholes with minimal diameter are placed in the sample. The connection is then created by melting the sample on the fiber inside a mirror furnace. Thereby, the Energy of $Q_{radiation}$ is used to melt the sample.

1.3. Experimental tensile calibration test setup

Two different experiments were conducted, which are both explained in this section. The results from the first experiment were used for calculations on the second experiment. The setup of the first experiment works as follows. For the first experimental setup, a special lab furnace and a tensile testing machine are coupled to apply strain to a cylindrical aluminum sample at different temperatures [5,25]. The principal setup is shown in Fig. 4. The universal testing machine has a movable cross beam (1) which divides the machine into two testing spaces, the upper (3) and the lower test space (5). The cross beam (1) is vertically actuated by two linear axes (6), which are powered by motors (9) and guided by a bearing. With the moving cross beam, tension or compression can be applied to different experimental setups. For this experiment, the mirror furnace (11) is equipped in the lower test space. Two tensile adapters (7) vertically placed are going through the furnace (11). A control unit (10) controls the devices and stores measurements, e.g., data from the interrogator measurements. The detailed view of the cross-section of the furnace shows the inside of the chamber. The ends

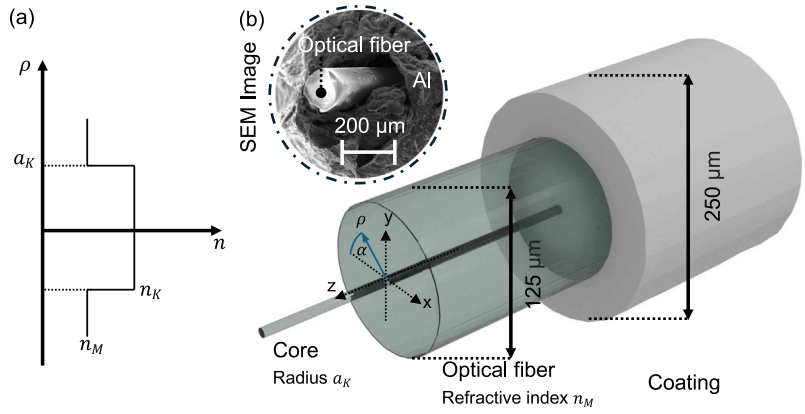


Fig. 2. Basic build of a single-mode fiber; (a), profile of the refractive index of a step indexed single-mode fiber [23]; (b), shell structure of the fiber from core to cladding [24], with image from Scanning Electron Microscope (*SEM*) at the interface of an aluminum specimen.

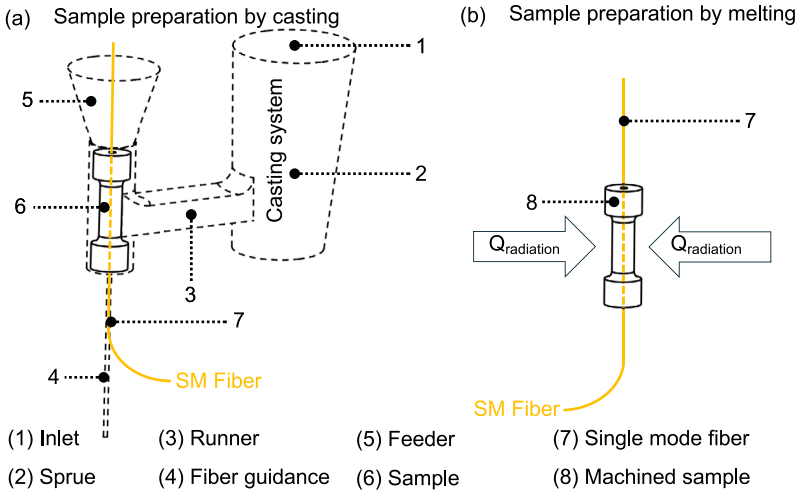


Fig. 3. Preparation schema of the calibration samples; (a), casting the samples; (b), melting the sample onto the fiber.

of the tensile adapters are screwed into titanium sleeves in which the cylindrical aluminum sample (13) is mounted. An isometric view of the sample (13) is shown in A. The sample vertically measures 14 mm and has a diameter of 6 mm. Because of the screw threads at both ends, the total axial length of the sample adds to 32 mm, and the diameter is increased because of the M10 screw threads at the end of the cylindrical specimen. The furnace (11) is equipped with four halogen lamps (15), each of which provides a power of 150 W [26]. The main body of this lab device consists of a cylindrical chamber, and in order to control the flow of the cooling water, water guards are installed. There is also an optical inspection via the safety glass possible. With the tension adapters (7), the sample (13) is connected via the two titanium sleeves (12) to the tensile testing machine. A high-temperature extensometer (2) helps to externally determine the strain and is used for validation. The extensometer is only visible on the background image in Fig. 4. It is equipped with two ceramic rods, which are placed on the edges of the titanium sleeves. In order to get a better understanding of the high-temperature tensile test, the detailed view in Fig. 4 shows a technical scheme of the sample geometry inside the furnace.

An important detail is that the aluminum samples (13) are cylindrical. The sensor is placed inside the test sample at the symmetrical center, see the position FBG (pos. FBG) highlighted in red, Fig. 4. The measurements with the interrogator can be taken directly in this region of interest because the reflective spectrum λ_{FBG} returns from the grating at this position back to the optical measurement equipment. The two arrows next to the sample indicate the direction of the applied strain ε_z. The described setup has the purpose of conducting calibration experiments to find calibration k-factors of κ. These are used for the real casting experiment, which works as follows.

1.4. Casting experiment with permanent mold

For the real casting experiments, a permanent mold is used to investigate hot cracking [27]. The mold is designed to have, on the one hand, hot spots where the melt cools down more slowly than in the rest of the casting, and on the other hand, consists of six cylindrical arms of different lengths. Each arm is a cylindrical rod and measures 10 mm in diameter. The greater the length of the arm, the higher will be the strain induced in this arm at the critical hot-spots. Because of the design of this permanent mold, it is called a star mold. Fig. 5 shows the experimental setup, in which (a) gives an overview of the lower permanent mold (1) and (b) shows the star geometry casting (6). The longest arm (2) of this geometry sits right opposite the shortest arm. This makes the positioning of a glass fiber (9) difficult, because the fiber would have to be guided by a very long capillary. This is why the fiber is placed at a shorter arm (3). This arm is better suited for the guidance of the capillary (8) and the fiber (9).

The mold's temperature is controlled by heating rods (4). The melt enters the geometry via a down sprue into the sprue basin (5), from which the arms are filled. For the experiments, one arm (7) is equipped with a glass fiber (9). The fiber is guided through a stainless steel capillary (8), and only at the position where the specific measurement

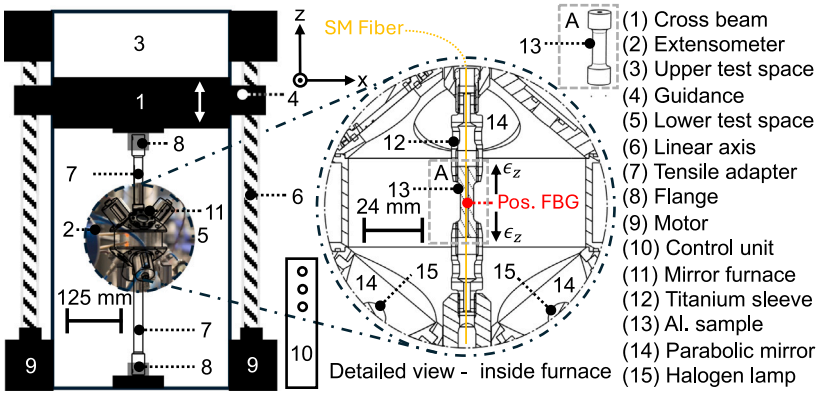


Fig. 4. Experimental setup of the tensile testing machine together with the mirror furnace and a representation of the samples.

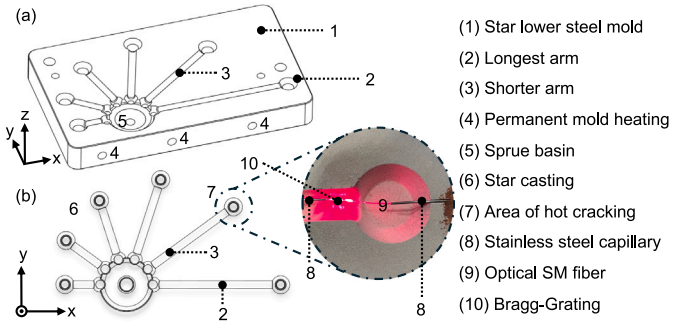


Fig. 5. Permanent star steel mold with different cylindrical arms and a detailed view on the placement of the optical single-mode fiber with the Bragg Grating.

is conducted, the fiber has contact with the melt (9), and the Bragg Grating (10) is directly exposed to the aluminum. The fiber is guided to the optical interrogator. The length of the fiber has nearly no influence, because the single-mode fiber can transfer the spectrum with minimal loss. The experiment is prepared by heating the permanent mold and pouring the liquid aluminum into the star mold geometry. The aluminum solidifies, and because of the shrinkage of the solid metal, the arms create different tensile strains. Because of the hot spots in this geometry, the aluminum solidifies later in the areas where there is more mass of aluminum. These areas have a possibility of hot tear formation.

2. Results

2.1. Experimental investigations

The experimental results are divided into two sections. First, the calibration results of the tensile test setup are explained, and second, the results from the real casting experiment of the star mold are represented. The results of the calibration experiments are directly used to conduct calculations on the star mold results.

2.1.1. Calibration experiment

Different experiments were conducted using the experimental setup of the tensile testing machine together with the mirror furnace. At first, the cylindrical sample is placed in the chamber of the oven and mounted to the tensile adapters. Then, a FBG is placed inside the sample borehole, which was placed at the center by drilling erosion. The hole has a diameter of approximately 0.3 mm. In the next step, the sample is heated up to 850 °C so that the aluminum liquefies and connects to the fiber, for the samples produced by melting. Because the temperature is way above the line of liquidus, the sample is stabilized with a graphite mold, which is afterward taken away. After this pre-processing of the sample, there is a connection to the fiber. The samples manufactured by casting are not heated up to this temperature since they are already connected to the fiber. Fig. 6 shows the wavelength shift of the first grating inside the fiber. The first curve was measured at room temperature (20 °C). The wavelength does not increase for this curve since, at the beginning, the temperature stays constant. The FBG's wavelength shifts as soon as the tensile testing machine starts to pull the sample in cyclic intervals, see Fig. 6 (ii) with increasing stress for each step, also see Fig. 7 for the exact values. Several peaks can be obtained with increasing height. After a certain number of cycles, the measurement is canceled to ensure that a specific force value (750 N) is not exceeded, preventing any damage from happening to the sample. After this, the furnace heats the sample to 100 °C; the signal curve is also plotted in Fig. 6. Here, the curve first shifts to a higher value. This is because of the increase in temperature in the sample, see Fig. 6 (i) heating. This increase is stronger for higher temperature values. The furnace holds the temperature at 100 °C. Then, the tensile testing machine starts to pull at the sample, which causes the rising peaks to occur, see Fig. 6 (ii) cyclic loading. Again, the tensile testing is cyclic, and the applied stress increases for each step. The same procedure was performed at 100 °C steps up to 500 °C. At 500 °C, the measurement started to show abnormal behavior, which is why no higher temperatures were applied to the sample. This behavior is discussed in the section of the discussion. In the end, the samples cool down, which leads to a drop in the signal, see Fig. 6 (iii) cooling. The strain and tensile force are measured by the tensile testing machine for each cyclic measurement. The result for the strain and the tensile force curves are shown in Fig. 7. The curves show the increase in the strain applied to the sample at the given temperatures. The increasing strain causes the force to increase to a value below 800 N. The curve's is different at each temperature. Generally, the tensile force decreases with higher temperatures, but the amount of strain increases. This effect is because the material behavior is temperature-dependent, and is called thermal softening. This effect is greatest at the highest temperature. For the calibration experiment, especially the strain applied is of interest since the FBG inside the sample is sensitive to this mechanical value. The shift of the wavelength is compared to the changes in strain. The strain is a calculated value from the tensile force measurement. It is more accurate because the measured strain by the extensometer is measured on the adapter sleeves. Figs. 8 and 9 show the respective curves focusing on the peaks for the 200 °C. The values of the peaks are given in both Figures and are determined by MATLAB's peak detection function.

With these two values of the strain and the wavelength shift and the following formula, the calibration factor (k-factor, κ) for each cycle can be determined, see Eq. (2). Where ε_z is the axial strain along the fiber

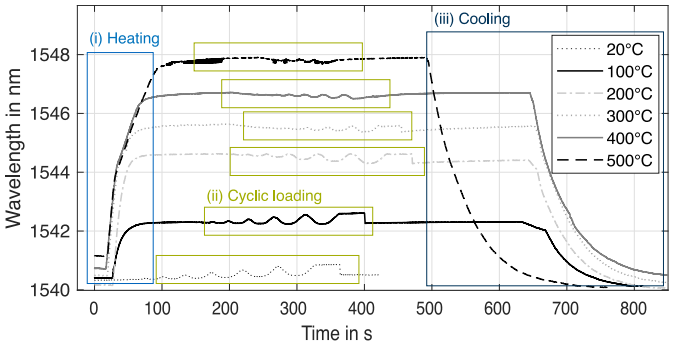


Fig. 6. Experimental results of the FBG inside the aluminum AlSi17 melted sample during (i) heating, (ii) cyclic loading, and (iii) cooling of the mirror furnace and the tensile testing for different temperatures.

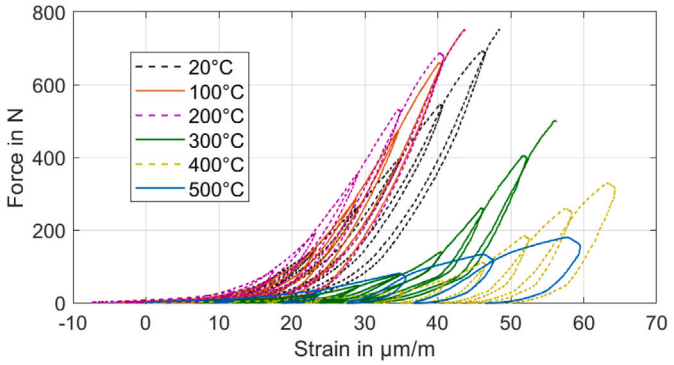


Fig. 7. Tensile force strain curves of the tensile testing machine at different temperatures.

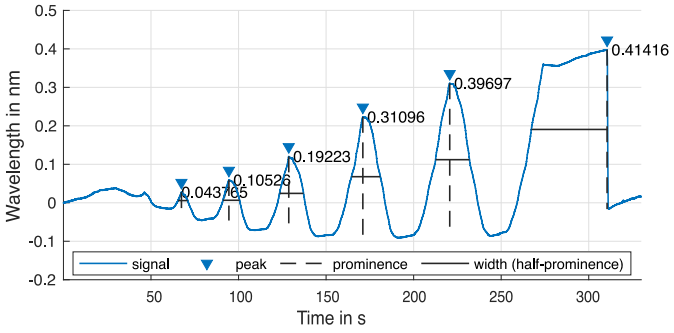


Fig. 8. Curve of the wavelength measurement of the FBG inside the cylindrical sample during tensile testing at 200 °C.

and $\Delta\lambda_{B,diff}$ is the shift of the wavelength of the fiber Bragg Grating, and $\lambda_{B,0}$ is the original reflective spectrum of the same Bragg Grating.

$$\frac{\Delta\lambda_{B,diff}}{\lambda_{B,0}} = \kappa\varepsilon_z \tag{2}$$

The resulting k-factors can be read from Fig. 10 for values at room temperature. Each box plot is calculated from one cyclic measurement. From the box plot, the minimum and the maximum value in the sample data can be read, represented as lower and upper bars. The 25th and the 75th percentile, which are the edges of the boxes, are represented as well as the median, which is the bar in the center. Any observations with a value more than 1.5 times the interquartile range away from the top or bottom of the box plot are defined as outliers [28,29]. The black box plot is the result achieved from the melted sample, which has a k-factor at room temperature of about 0.68. The light blue and

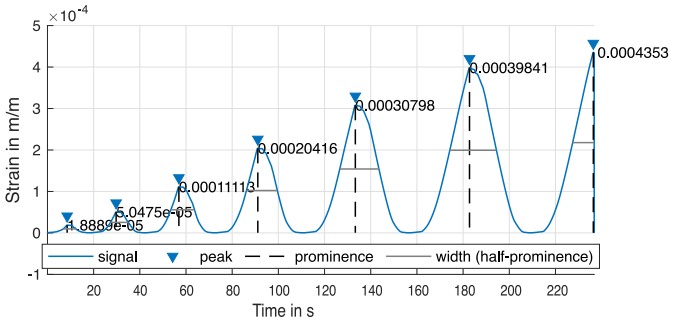


Fig. 9. Showing the strain calculated from the tensile force measurement during the cyclic tensile test at 200 °C.

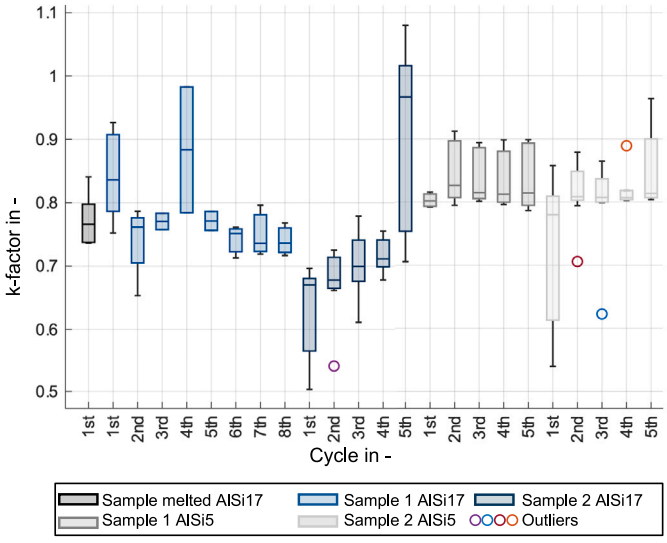


Fig. 10. Graph showing the k-factor values as box charts, where one box chart represents one cyclic measurement at room temperature.

Table 2
List of mean k-factors calculated from different samples.

Sample color	Alloy	Melted = M. Casted = C.	Mean k-factor	Sample size n
Black	AlSi17	M	0.68108	8
Light blue	AlSi17	C	0.79026	28
Dark blue	AlSi17	C	0.71351	38
Dark gray	AlSi5	C	0.83537	25
Light gray	AlSi5	C	0.79314	38

dark blue box plots give the results for the hyper-eutectic AlSi17 alloy. Whereas the gray and the light gray box plots represent samples made of the hypo-eutectic AlSi5 alloy. Only the sample colored in black is prepared by melting the fiber and the alloy together to create bonding. All other samples are manufactured by casting.

The exact k-factors at room temperature are listed in Table 2. The mean values vary between the minimum value of 0.68 and the maximum value of 0.83. Showing no significant difference between the casted and the melted samples at room temperature. Comparing the different alloys with each other, there is a much lower scatter in the data for the AlSi5 compared to the AlSi17. Overall, the k-factors are in the interval between 0.5 and 1.1, which is also the result of the state of the art.

For the higher temperatures, the k-values are represented in Fig. 11. The range of the k-factors is now increased compared to the results at room temperature. The values are also not distributed as evenly as before, which is specified by a closer analysis later in Fig. 12. This

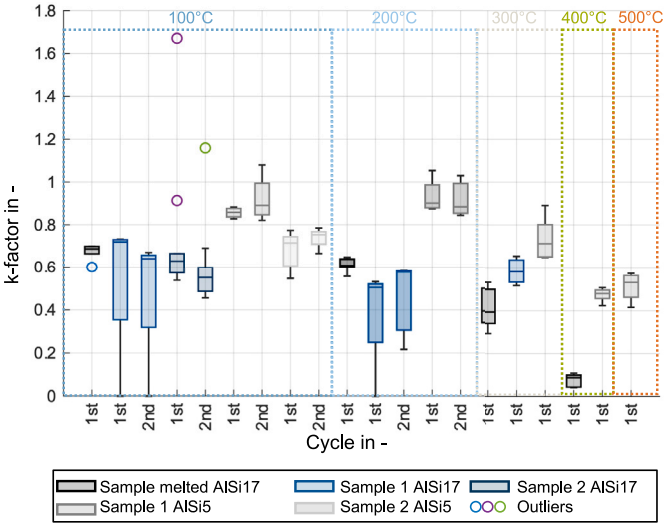


Fig. 11. k-factor values as box charts, where one box chart represents one cyclic measurement at higher temperatures.

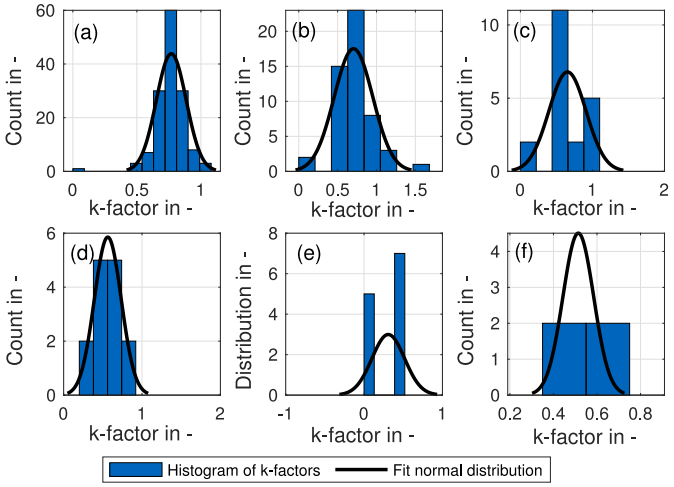


Fig. 12. Different distributions of the k-factor measurements; (a), at 20 °C; (b), at 100 °C; (c), at 200 °C; (d), at 300 °C; (e), at 400 °C; (f), at 500 °C.

creates a greater uncertainty in the determination of the k-factors. With rising temperature, fewer samples give valid data because the signal of the FBG decays or is suddenly lost during measurement. This happens because the fiber is exposed to more extreme conditions at higher temperatures and cannot always withstand them. Only for the AlSi5, a valid measurement at 500 °C is obtained, resulting in a k-factor slightly lower than the one at room temperature. For the AlSi17, only the melted sample survived the calibration testing to a temperature of 400 °C. The resulting k-factor has the lowest mean value at this temperature. The higher the temperature, the lower the mean value of the melted sample becomes. This trend can also be seen with the gray AlSi5 sample as well as with the light gray sample if the values from room temperature are also considered for comparison.

To discuss the experimental results, it is necessary to look closely at the errors because measurements can vary. Determining and specifying the corresponding errors helps to analyze the data's reliability. Different sources of errors exist, which can add up. The signal can be noisy for optical measurements if light from two different optical devices of the same bandwidth is interacting with each other. But still, random measurement deviations can occur, and these have to be estimated. [30]

Thus, looking at the data distribution helps to make a statement of the statistical confidence [31]. Therefore, we first look at the data distribution of all the k-factors at 20 °C and compare it to the distribution of the values at 100 °C and up to 500 °C. Fig. 12 shows the distribution of the k-factors at different temperatures and the fit of the normal distribution.

The fit shows how close the data sets are to a normal distribution. Apart from Fig. 12 (e), where the data is more bimodal-like distributed, the rest of the figures show a more Gaussian-like distribution as expected. For Figs. 12 in the second row (d), (e), and (f), there is also the problem that the sample size is comparatively low. The exact values of the sample size for each data set are listed in Table 3. The sample size n of the measurements decreases with rising temperatures. This is simply because of the complexity of the experiment at higher temperatures. Table 3 lists several statistical measurements to specify the results even more. These are the mean, median, standard deviation, and the confidence interval of the error with a statistical safety of 95%. The mean value is useful if the data does not contain too many statistical outliers. This may also be true for other measures of central tendency, but the median is a little more robust against such outliers.[32] Which is why, for the values in Fig. 12 (a) and the corresponding values in Table 3, the median is slightly higher than the mean. This is because the median is not so strongly affected by the outlier. The data's variability is given with the standard deviation and the confidence interval. The standard deviation gives the mean distance of all k-factor values from the mean. It is lowest for the values at room temperature, and interestingly, even lower for the highest temperature. This shows that for 500 °C, the measures that were conducted were very close to each other, but the small sample size influences this fact. Therefore, looking at the confidence intervals is also important. With 95% certainty, the confidence interval gives the lower and upper limits of the k-factors lying within this range. The upper limit of the confidence interval seems to be lower for temperatures above 300 °C. This could indicate that the k-factors decrease slightly with higher temperatures. However, for the highest temperature value of 500 °C, the confidence interval is greater because the sample size is so small, making it more complicated to compare against.

2.1.2. Hot tearing test

Since Sample 1 of the AlSi5 survived the calibration process up to 500 °C, this sample was used to investigate hot tearing in situ. Therefore, the sample is heated above the solidus temperature of up to 585 °C. This temperature is maintained, and the sample is torn by tensile force in the axial direction. Fig. 13 shows the in situ FBG signal collected during the event of tearing and the ex situ signal of the tensile testing machine. Both signals indicate the strain that occurred and led to a hot crack. For the FBG measurement, the k-factor of 0.5142 from the highest calibration result of the experiment is taken. The temperature-dependent Young's modulus is considered to be about 1.1 GPa. The so calculated strain results give an indication that hot tearing around the line of solidus starts as soon as the strain reaches values close to 4e−3, and a crack separating the material and consequently causing a loss of the FBG signal occurs at a strain above 5e−3.

The results of the calibration experiment show a set of k-factors for different aluminum alloys at high temperatures. For the AlSi5 hypo-eutectic alloy, a hot tear criterion can be defined, stating that tearing occurs as soon as the strain is above 4e−3 at around the solidus temperature. Because this criterion is only defined for the hypo-eutectic alloy, the following star mold experiment was conducted only with this specific aluminum composition.

Table 3
List of statistical values describing the distribution of the measured calibration values of the k-factor.

k-factor at diff. temps.	Mean	Median	Standard deviation	Lower confidence interval (95%)	Upper confidence interval (95%)	Sample size
T	M_{mean}	m_{median}	σ_{std}	z_{low}	z_{up}	n
20 °C	0.7685	0.7866	0.1163	0.749219	0.787809	142
100 °C	0.70425	0.6893	0.248721	0.635005	0.773494	52
200 °C	0.653524	0.6088	0.258527	0.532529	0.774519	20
300 °C	0.563168	0.5584	0.171681	0.464042	0.662294	14
400 °C	0.308795	0.4395	0.207833	0.176744	0.440846	12
500 °C	0.514182	0.5323	0.0707584	0.40159	0.626775	4

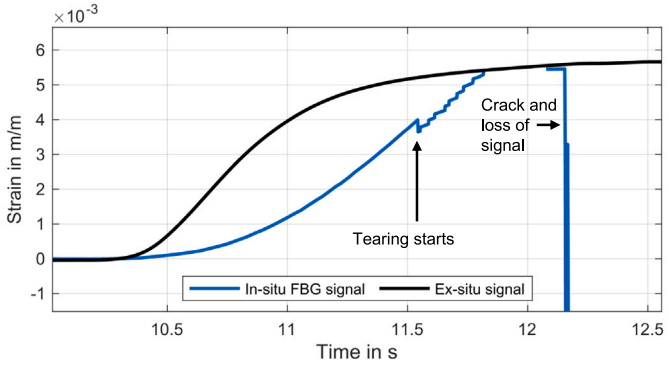


Fig. 13. Curves of the in situ and ex situ measurement of hot tearing inside an AlSi5 sample.

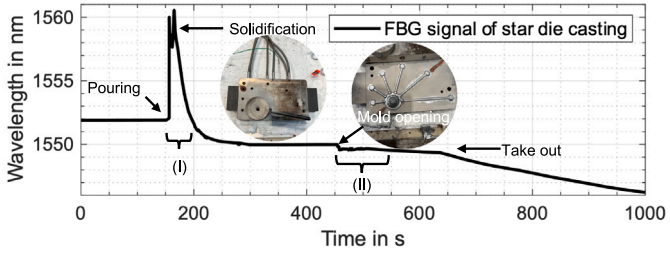


Fig. 14. Permanent star steel mold that provokes hot cracking.

2.1.3. Star mold casting experiment

The experiment was conducted using the described star permanent mold. The wavelength of the FBG was measured at 10 Hz to monitor the interaction of the process with the fiber. The signal curve is shown in Fig. 14 as a black line. Until 150 s, the signal is constant and has a value between a minimum of 1551,8866 nm and a maximum of 1551,9066 nm.

At this point, the signal had already drifted from its original reflective wavelength ($\lambda_{B,0}$ = 1550.03 nm) because the mold had a temperature of 180 °C. The temperature of the melt was measured to be about 780 °C. After 156 s, the signal immediately rises to a first peak; here, the permanent mold was filled with the aluminum melt, which causes the temperature to increase rapidly at the FBG. Then, a second peak forms; this first phase (I) is between 155 s and 190 s, is shown in detail in Fig. 15, and is discussed in the next paragraph. After the second peak, the casting cools down, even below the original wavelength, and converges to a steady value between a minimum of 1549,9716 nm and 1549,9891 nm. At this point, the temperature is no longer decreasing, and the casting is solidified. After this constant phase, there is a substantial drop in the signal at the second phase (II); this is because the mold was opened, leading to a change in the signal behavior. After this second phase, there is a slight change in the gradient of the signal. At this point, the casting was completely removed from the mold, which led to a higher cooling rate. For the star mold experiments at specific points, the formula (2) applies. Some approximations are made to satisfy the problem of cross-sensitivity. For both calculations, the assumption of neglecting any temperature changes at the fiber is made at Phase I and Phase II. This assumption can be made because the jumps in the signal happened immediately, and temperature changes can only happen over a certain amount of time. Only in the special case of pouring an immediate change of wavelength occurs due to the liquid melt hitting the Bragg Grating. But for Phase I and Phase II, the fiber is already surrounded by aluminum. Therefore, changes in temperature do not happen that rapidly anymore. In order to calculate the changes of the state of strain ε_z for these two cases, the value of k-factor is $\kappa_{@roomtemp}$ = 0.751 [18] and the mean value of the k-factor $\kappa_{mean,@200\ °C}$ for Phase II and $\kappa_{mean,@500\ °C}$ for Phase I, from the experimental result of the calibration experiment are compared. First, we look at Phase I. Fig. 15 shows a detailed view of Phase I. Two significant events take place during this time of the processing. The first leads to the first peak, which has a higher gradient, as can be seen in the first derivative. This peak is the moment where the melt touches the fiber. This leads to the immediate shift of the wavelength of the signal. After this peak, the solidification starts with a classic solidification behavior. There is at first a high rate of cooling, i.e. the first derivative quickly drops, and as soon as the solidification starts, the curve flattens because of the latent heat. Shortly after, a second peak starts to build. The wavelength again shifts in the positive direction. This point is marked with an arrow pointing at this position, indicating the value of λ_{t1}. The peaks come to an end at the point of λ_{t2}. This second peak of the signal clearly indicates that there is either a very high increase in temperature or a high increase in tensile force. Looking at the first peak, it can be assumed that at the position of the FBG the melt has already gone through the line of liquidus towards the line of solidus. This gives reason for concluding that high tensile forces are pulling at the fiber in the axial direction, provoking hot cracking, which is exactly the aim of the star mold. Therefore, it is of interest to calculate the stress, which occurs at Phase I. Table 4 lists the input values. For comparison, the calculations are done with two different k-factors.

The first k-factor $\kappa_{@roomtemp}$ is a value from literature. The second k-factor $\kappa_{@500\ C}$ comes from the described calibration experiment and is the mean value over the values determined at 500 °C for the alloy AlSi5. The values of the wavelength come from the measurement or are values from the data sheet. The following Eq. (3) is the base for the calculation.

$$\frac{\Delta\lambda_B + \Delta\lambda_{B,th}}{\lambda_{B,0}} = \kappa\varepsilon_z \tag{3}$$

The Eq. (3) is simplified to, see Eq. (4):

$$\frac{\Delta\lambda_{t12}}{\lambda_{B,0}\cdot\kappa} = \varepsilon_z \tag{4}$$

This is because we neglected the influence the temperature had to the peak and assumed that it did not have a significant influence. This

Table 4
List of input values for Phase I and Phase II.

	Unit	Phase I		Phase II	
		$\kappa_{@\text{room temp}}$	$\kappa_{\text{mean},@500°C}$	$\kappa_{@\text{room temp}}$	$\kappa_{\text{mean},@200°C}$
k-factor, κ	(–)	0.751	0.5142	0.751	0.9274
Wavel., λ_{t1}	(nm)	1.55763e+03	1.55763e+03	1.54998e+03	1.54998e+03
Wavel., λ_{t2}	(nm)	1.56054e+03	1.56054e+03	1.54955e+03	1.54955e+03
Wavel. diff., $\Delta\lambda_{t12}$	(nm)	2.905396	2.905396	0.4313	0.4313
Length, L	(mm)	3.7	3.7	3.7	3.7
Young's mod., E	(GPa)	70	70	70	70
Young's mod., E_{temp} [a]	(GPa)	70	29	70	63
Cross-sec. area, A	(mm^2)	78.539	78.539	78.539	78.539

[a] Material properties of the Young's Modulus E_{temp} are estimated values based on aluminum databases from the Simulation Software Magmasoft and do not claim to be absolutely precise. For Phase I, the E_{temp} is taken for a temperature at 500 °C, for Phase II it is assumed for a temperature at 200 °C.

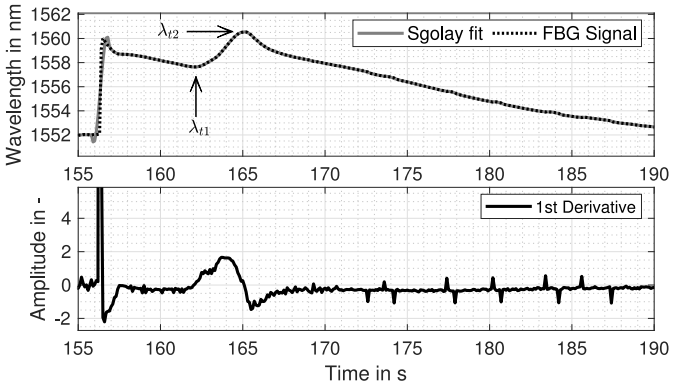


Fig. 15. Curves of Phase I, with the corresponding first derivative.

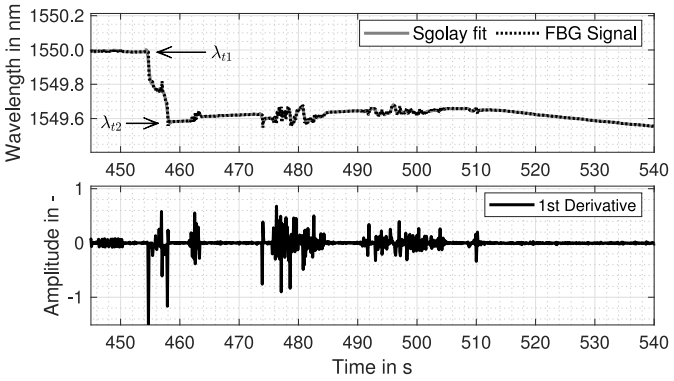


Fig. 16. Showing the curves from Phase II with the results of the first derivative.

causes an error, but we can find the value of the strain ε_z. The value of ε_z is given in Table 5. With this value and Eq. (5):

$$\varepsilon = \frac{\Delta L}{L} \tag{5}$$

The values of ΔL are calculated, see Table 5, and also the stress and the force leading to this peak are calculated by the following two Eqs. (6) and (7), where A is the cross-sectional area with radius $r = 3$ mm.

$$\sigma = E \cdot \varepsilon \tag{6}$$

$$F = E \cdot \varepsilon \cdot A \tag{7}$$

After Phase I, the signal decreases again because of further cooling. At this point, the first derivative shows repeating peaks. This is because the FBG is exposed to harsh conditions, which is why the amplitude of the signal sometimes splits into two peaks. This second peak makes it hard for the software of the interrogator to detect the actual wavelength of the reflective spectrum, which leads to the jumps observed in the first derivative in Fig. 15. Next, we look at the second Phase II. At this point, there is another immediate change in the signal. At this point, the cooling rate of the casting had already converged, because there was no significant change in the signal for a certain amount of time. From temperature measurements, it is documented that the temperature of the permanent mold is around 200 °C at this point. Because the temperature can again be assumed to have no influence on the signal jump, another calculation with the already introduced formulas is made. The input values for the second Phase II results can again be read from Table 4. This time the two k-factors $\kappa_{@roomtemp}$ and $\kappa_{@200\ C}$ are compared. The values of the λ_{t2} and λ_{t2} were measured for the maximum and minimum value for this peak as indicated in Fig. 16. This peak was mainly the result of the opening of the upper part of the star mold. This caused the drop in the signal, which can be interpreted as a shrinkage of the geometry, which makes sense because by removing the mold, the casting is free to shrink. The results of this shrinkage are calculated and listed in Table 5. This event was smaller but also interesting for this investigation.

Comparing the different process events, the behavior of the signal matches the process during Phase II very well and makes sense from a technical view. However, the results calculated for Phase I are high. Therefore, the temperature-dependent Young's modulus should be taken into account. For the AlSi5 alloy, the value is assumed to be approximately 29 GPa for 500 °C. Due to this fact, the results for the stress and the force are more or less half of what they are with the Young's modulus at room temperature, which is more reasonable, see Table 5. But looking at the strain, which is already below 0.4e−3 from the hot tear calibration experiment, we can conclude that this was just not enough strain to force hot tearing to occur. Also, this result corresponds to the observations we made when we optically investigated the casting, because no significant appearances of tears were seen. Also, the signal of the FBG is maintained till the end of the process; if the star mold had torn, the fiber would also have ripped, which would have caused a loss of signal, just as in the experiment in the furnace.

2.2. Numerically computed results

The star mold's behavior is investigated using a hypo-eutectic alloy system, AlSi6(Cu4), and a hyper-eutectic alloy system, AlSi17(Cu3), for comparison. The material database used for these alloys closely resembles the one utilized in the experiments, allowing for an approximation in thermal and strain simulations to validate the distribution of these parameters. The star mold is modeled with the same mesh, consisting of 2038608 Cartesian cells, and the evaluation region is identical. Since the pouring process influences the temperature distribution, this was simulated first to enhance the accuracy of the results. Consequently,

Table 5
List of the values calculated from the input values.

	Unit	Phase I		Phase II	
		$\kappa_{@\text{room temp}}$	$\kappa_{\text{mean},@500°C}$	$\kappa_{@\text{room temp}}$	$\kappa_{\text{mean},@200°C}$
Strain, $\Delta\varepsilon_z$	(m/m)	2.495955e−3	0.0036	3.7050e−04	3.0004e−04
Length, ΔL	(μm)	9.23	13.4218	1.0±	±1.1102
Stress, $\Delta\sigma$	(MPa)	174.7168	253.9259	25.9350	21.0031
Stress, $\Delta\sigma_{temp}$	(MPa)	–	105.1978	–	18.9028
Force, F	(kN)	13.722	19.943	1.9	1.649
Force, F_{temp}	(kN)	–	8.2621	–	1.4841

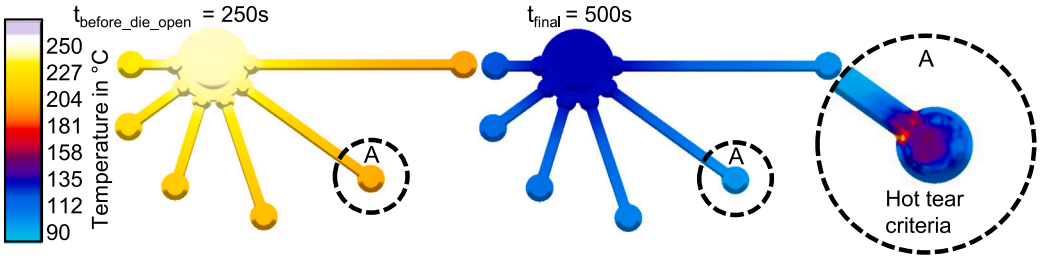


Fig. 17. Simulation results of the temperature distribution of the star casting and a detailed view of Magma's hot tear criteria at one of the arms of interest.

the outcome of the pouring process becomes the initial temperature distribution from which solidification is calculated. But the results of the pouring process do not affect the final findings, which is why Fig. 17 focuses solely on the calculated solidification. This figure presents the overall results of the temperature before opening the die and at the final time step. Also, the result of the hot tear criteria, which is a susceptibility, calculated by Magmasoft, is represented in a detailed view A in Fig. 17. Because of the result of the hot tear criteria, the FBG is placed in this region. It shows that this region is most likely to experience hot tearing. For the temperature results, generally, the global results show that the longer arms of the star cool down faster than the shorter arms. Because the FBG measurement is not only dependent on the temperature but also on the stress occurring along the arms, the global results of the tensile strength of the star mold are represented in Fig. 18. Again, we are looking at two time steps to get a global overview of the distribution of the tensile strength results of the simulation. Both results show that the values of stress are higher for the longer arms when compared to the shorter ones. Therefore, the risk of tearing generally increases with longer arms. However, with the FBG, it is only possible to conduct measurements locally. For this reason, the region where the FBG is placed in the experiment is also evaluated more closely in the simulation. To compare the experimental measurements more easily with the numerical results, the simulation's numerical temperature and mechanical strain results are recalculated into wavelength values with the following Eq. (8) and Eq. (9). The thermal expansion coefficient of the fiber in the axial direction is described by α, and the thermo-optical coefficient is given by η. The value of $T_{Simulation,result}$ is the theoretical value calculated from the simulation of the solidification. The parameters used for the calculation are given in Table 6 and depend on the used glass fiber.

$$\Delta\lambda_{\text{Temp,FBG}} = (\alpha + \eta) \cdot \lambda_{\text{FBG0}} \cdot T_{Simulation,result} \tag{8}$$

For Eq. (9) n_{eff} is the initial refractive index and p_{11} and p_{12} elastic optical coefficients. The value of $\varepsilon_{\text{Simulation, result}}$ is calculated from the simulation and describes the mechanical strain occurring during solidification caused by the shrinkage of the cooling alloy.

$$\Delta\lambda_{\text{Strain,FBG}} = \lambda_{\text{FBG0}} \cdot \left(1 - \frac{n_{\text{eff}}^2}{2}\left(p_{12} - \nu \cdot (p_{11} + p_{12})\right)\right) \cdot \varepsilon_{\text{Simulation,result}} \tag{9}$$

Fig. 19 shows the calculation's resulting curves at the specific evaluation region over time. The qualitative curve course and the scaling of the values of the wavelength reflect the experimental data. In the curve of the temperature, as well as in the curves of the tensile strain

Table 6
List of theoretical optical parameters used for the numerical calculation.

Parameter	Value	Unit
α	0.52e−6	–
η	7.58e−6	–
λ_{FBG0}	1547.5	nm
n_{eff}	1.45	–
p_{12}	0.252	–
p_{11}	0.113	–
ν (Poisson number)	0.16	–

results, there are specific time steps where the course of the curve shows a very similar behavior compared to experimental results. The take out (Phase II) especially creates a similar result in the simulation. In the first phase during the solidification, we also see an increase in the strain, see also Fig. 20 for a more detailed view. The results were evaluated for the hypo-eutectic alloy and the hyper-eutectic alloy. The curves differ at some points but still show a similar behavior over time. This indicates that the difference between the alloys is not that great, but for the creation of hot tears, this cannot be concluded from the simulation. A difference can be investigated in the strain results in the first phase, see Fig. 20. There is a little increase in the tensile strain, as measured in the real experiment, but this increase is much lower than in the casting experiment. This is because it is more difficult for the simulation to calculate strain at temperatures around the solidus temperature. In this case, the numerical simulation and the experiment follow the same trend but with different magnitudes. The choice of the alloy has again not much influence on this general result.

3. Discussion

For the discussion, we look at three different aspects of this paper. First, we discuss the importance of the correct k-factor κ value; second, we discuss the calibration method approach; third, we discuss the casting experiment. The k-factor κ is the primary factor influencing error in determining stress because it has a large weight in the mathematical model. Using just slightly different k-factors, the result changes strongly compared to other factors. Therefore, the reliability of the k-factor is essential for the outcome of the result. Since the literature already presents different k-factors, it is not easy to decide which one to trust. The current approach from the state of the art assumes that

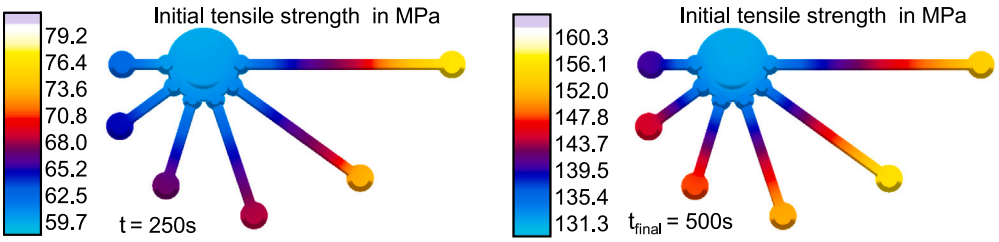


Fig. 18. Simulation results of the tensile strength distribution during the solidification of the star casting.

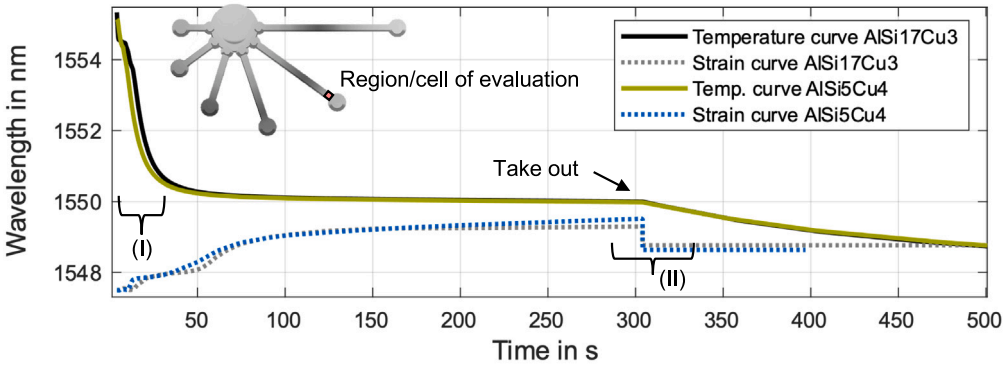


Fig. 19. Simulation results of the star mold showing the calculated theoretical wavelength curves during solidification at the region of evaluation.

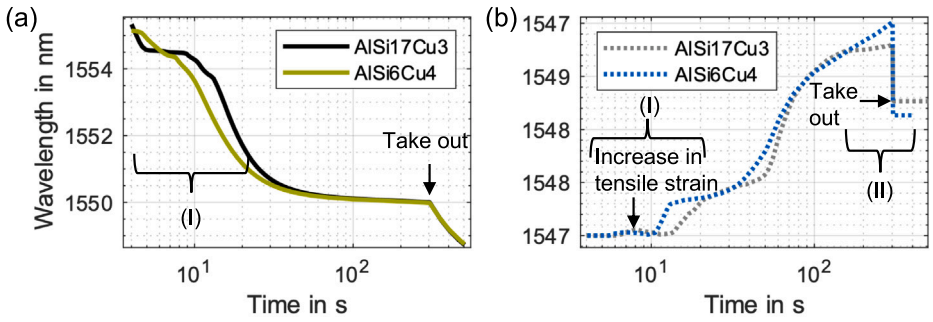


Fig. 20. Detailed view of the simulation result curves from the region of evaluation.

the k-factor is a constant and is only valid for experiments at room temperature. The calibration experiment gives reason to assume that the k-factor has some kind of temperature-dependent behavior. In our case, the k-factor had a tendency to decrease for higher temperatures. This would mean that the shift of wavelength caused by strain is higher for higher temperatures compared to the real strain. This gives rise to the assertion that FBGs become more sensitive to strain with rising temperature. One assumption could be that with higher temperature, the fiber's Young's modulus decreases, leading to a greater sensitivity towards forces acting on it. However, this behavior must be considered when choosing a k-factor. This paper showed an approach to gain more information about the temperature-dependent behavior of the k-factor. The collected data helps to make a decision on the right k-factor use. The methods of melting and casting samples are both valid approaches, but samples made from casting are advised to be used. We experienced many hurdles working with the melted samples and had only one that gave good signal curves. Other samples had problems in bonding during the melting process, and the fiber repeatedly slipped out of the sample because there was not enough friction between the compounds. We further investigated the adhesion between the fiber and aluminum. Fig. 21 shows the problem that occurs when the fiber is placed inside a pre-drilled hole of the aluminum sample, when melted onto the fiber instead of casting around the fiber. Using this approach creates many defects globally. The scattering electron microscope images on the surface of the sample where the fiber exits the sample already show that the fiber is not embedded perfectly, see Fig. 21 (a).

There are some interfaces with adhesion between the fiber and the aluminum. Still, the bonding often struggles to reliably transfer the strain applied to the aluminum sample to the fiber. Also, the micrographs in Fig. 21 (b) show that the fiber is embedded inside the sample, but many defects cause a weak bonding between the fiber and the aluminum. Nevertheless, the micrographs also show areas such as the detailed view in (b) A, where the connection was better than in the detailed view B. If we look at the process of how the aluminum is melted onto the fiber, we can suspect that aluminum oxides must build up at the surface of the borehole. The oxides create a dividing interface between the aluminum and the fiber. Therefore, the quality of the results is believed to increase, just as oxide films resist bonding in aluminum casting and lead to cracks [33]. In comparison, during casting the aluminum around the fiber, the liquid melt wets the fiber, and the aluminum oxides are swept to the feeder. Therefore, we also investigated the bonding of one sample, which was manufactured by casting. Fig. 22 shows two micrographs from different cutting planes along the cylindrical specimen. As expected, some normal porosity is

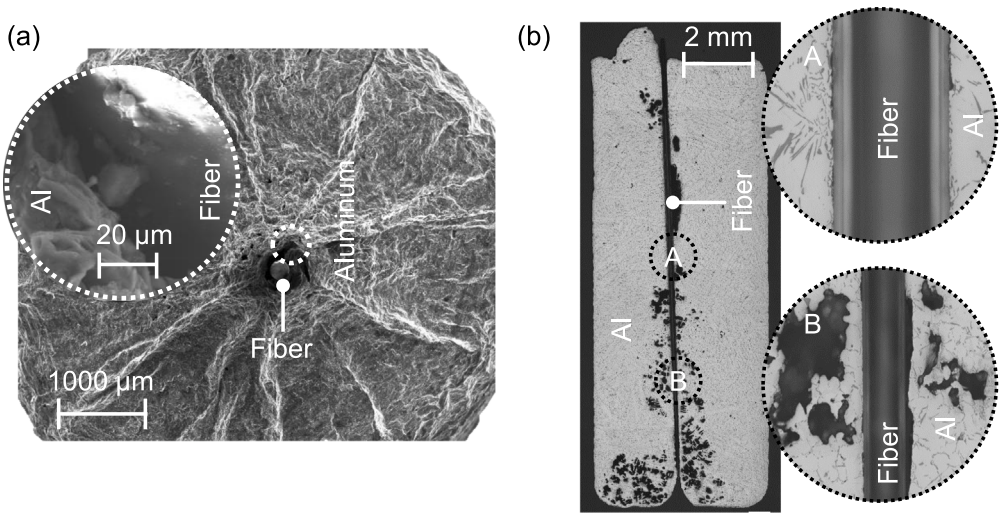


Fig. 21. Different cut-outs of optical inspections; (a) SEM Images of a cylindrical aluminum sample with a fiber melted at the radial center; (b), micrographs of the sample showing the fiber position and the connection to the aluminum.

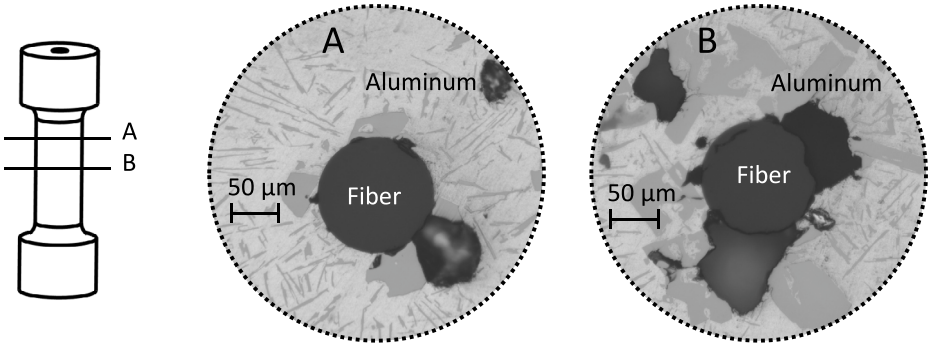


Fig. 22. Different cutting planes along a sample made by gravity casting.

visible around the fiber, which every casting has, but especially the cutting plane A, which is mostly connected to the fiber. However, the cutting plane B shows more porosity surrounding the fiber. Still, along the axial axes, the bonded areas sum up to create a decent connection for the strain to be transferred between both materials.

At this point, we discuss the results of the actual casting experiment. For the casting experiment, we looked at two different states of the solidification process. The first phase was the phase in which the solidification just started. Therefore, a higher-value k-factor and the k-factor from the literature were compared. For easier comparison, the values of the Young's modulus were constant but also temperature dependent. For the case of temperature-dependent Young's modulus at 500 °C, the calculated values are much lower. The choice of Young's modulus leads to an error when the resulting stress is calculated. Nevertheless, the stress calculated for Phase I is in a range where it makes sense to have this amount of stress. The same is true for the second Phase II. Here, the permanent mold was opened, which is why there was a shift from a higher wavelength to a lower wavelength. The release of pressure stress was approximately between 18.90 and 25.93 MPa. These values can quickly build up in castings when they are solidified in permanent molds because of the hindrance to the shrinkage. Problems that usually occur in a casting geometry, like the star mold, are that the positioning of the fiber is challenging. Especially when filling the mold, the fibers can easily be washed away by the aluminum melt. For the casting experiment, this did not happen because most likely, fibers tend to break when they move too much.

4. Conclusion

The results show that calibration samples manufactured by casting are more durable and have good bonding between the fiber and the alloy.

Evaluating the k-factors is a conquering task, and many errors can influence the result. Especially the connection between the fiber and the aluminum influences the result. Therefore, we recommend using the process of melting the aluminum to the fiber only if there is no other way of placing the fiber inside the sample, or if only the temperature measurement is of interest. Better results can maybe be achieved by the use of flux. In the case of tensile testing, the adhesion of the fiber inside the aluminum is very important. Therefore, casting the sample around the fiber creates a stronger bonding with less porosity. Also, the aluminum oxides do not build a separating layer. Future samples will therefore be prepared using the approach of casting the tensile test geometries and with a casting setup that follows basic casting rules, see [34]. Nevertheless, the tensile testing setup is an experimental setup that works and can be used for calibration experiments, since it is possible to maintain a constant temperature of the sample and conduct different loadings to the sample during the temperature state. Moreover, we showed that with a sample made by casting, we are even able to in situ measure hot tearing for the hypo-eutectic AlSi5 alloy.

CRediT authorship contribution statement

Constantin Bauer: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Robert Kleinhans:** Writing –

review & editing, Resources, Investigation. **Maximilian Erber:** Visualization. **Manuel Pintore:** Writing – review & editing, Resources, Investigation, Data curation. **Christoph Hartmann:** Supervision. **Wolfram Volk:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

A special thanks to the German Research Foundation (*DFG*) for supporting this research and for providing measurement equipment, such as the interrogator under grant 456662088.

Data availability

Data will be made available on request.

References

[1] K. Weraneck, F. Heilmeier, M. Lindner, M. Graf, M. Jakobi, W. Volk, J. Roths, A.W. Koch, Strain measurement in aluminium alloy during the solidification process using embedded fibre bragg gratings, Sensors (Switzerland) 16 (2016) http://dx.doi.org/10.3390/s16111853.

[2] S. Khadka, P. Predecki, M. Kumosa, J. Hoffman, Monitoring solidification of tin-bismuth alloys using fbg sensors, Materialia 21 (2022) http://dx.doi.org/10.1016/j.mtla.2022.101320.

[3] F. Heilmeier, R. Koos, K. Weraneck, M. Lindner, M. Jakobi, J. Roths, A.W. Koch, W. Volk, In-situ strain measurements in the plastic deformation regime inside casted parts using fibre-optical strain sensors: A contribution to the production of intelligent parts, Prod. Eng. 13 (2019) 351–360, http://dx.doi.org/10.1007/s11740-019-00874-7.

[4] Q. Bian, A. Podhrazsky, C. Bauer, A. Stadler, F. Buchfellner, R. Kuttler, M. Jakobi, W. Volk, A.W. Koch, J. Roths, Temperature and external strain sensing with metal-embedded optical fiber sensors for structural health monitoring, Opt. Express 30 (19) (2022) 33449–33464.

[5] C. Bauer, M. Erber, G. Fuchs, T. Brügge, C. Hartmann, W. Volk, Analysis of the melting and solidification process of aluminum in a mirror furnace using fiber-bragg-grating and numerical models, IOP Conf. Ser.: Mater. Sci. Eng. (2023) 1281, http://dx.doi.org/10.1088/1757-899x/1281/1/012063.

[6] L. Meier, M. Hofmann, P. Saal, W. Volk, H. Hoffmann, In-situ measurement of phase transformation kinetics in austempered ductile iron, Mater. Charact. 85 (2013) 124–133, http://dx.doi.org/10.1016/j.matchar.2013.09.005, URL https://www.sciencedirect.com/science/article/pii/S1044580313002581.

[7] J. Mills-Brown, K. Potter, S. Foster, T. Batho, The development of a high temperature tensile testing rig for composite laminates, Compos. Part A: Appl. Sci. Manuf. 52 (2013) 99–105, http://dx.doi.org/10.1016/j.compositesa.2013.04.009, URL https://www.sciencedirect.com/science/article/pii/S1359835X13001206.

[8] Z. Guo, X. Wang, Y. Liu, Y. Liu, F. Li, Longitudinal tensile test on weld metal in fusion zone at high temperature, J. Constr. Steel Res. 172 (2020) 106174, http://dx.doi.org/10.1016/j.jcsr.2020.106174, URL https://www.sciencedirect.com/science/article/pii/S0143974X19315469.

[9] Y. Guo, F. Qi, Y. Wang, L. Zhang, S. Pang, Q. Li, G. Zeng, J. Zhan, Q. Li, Y. Wang, L. Guo, G. Wu, An extended criterion for hot cracking susceptibility of al-li alloys, Eng. Fail. Anal. 171 (2025) 109348, http://dx.doi.org/10.1016/j.engfailanal.2025.109348, URL https://www.sciencedirect.com/science/article/pii/S1350630725000895.

[10] L. Katgerman, D. Eskin, In Search of the Prediction of Hot Cracking in Aluminium Alloys, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 11–26, http://dx.doi.org/10.1007/978-3-540-78628-3_1, Ch. unknown.

[11] A.H. Fritz, G. Schulze, Urformen, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 5–114, http://dx.doi.org/10.1007/978-3-540-76696-4_2.

[12] Failure Analysis Failures Caused By Thermal Loading Vdi 3822 Blatt 1.4 / Part 1.4 Vdi-Handbuch Werkstofftechnik Vdi-Handbuch Fabrikplanung Und-Betrieb, Band 1: BetriebsÜberwachung/Instandhaltung Vdi-Handbuch Produktentwicklung Und Konstruktion Vdi-Handbuch Verfahrenstechnik Und Chemieingenieurwesen, Band 3: VerfÜgbarkeit/Schadensanalyse, Technical Report, 2011, URL www.vdi-richtlinien.de.

[13] C. Ma, R. Zhang, Z. Li, Y. Wang, C. Zhang, H. Yin, X. Qu, Effects of solidification shrinkage on solute segregation and hot cracking sensitivity in liquid channel during columnar dendrite growth, J. Mater. Res. Technol. 31 (2024) 2367–2375, http://dx.doi.org/10.1016/j.jmrt.2024.07.008, URL https://www.sciencedirect.com/science/article/pii/S2238785424015618.

[14] L. Li, R. Zhang, Q. Yuan, S. Huang, H. Jiang, An integrated approach to study the hot tearing behavior by coupling the microscale phase field model and macroscale casting simulations, J. Mater. Process. Technol. 310 (2022) http://dx.doi.org/10.1016/j.jmatprotec.2022.117782.

[15] K. Kuang, W. Cantwell, Use of conventional optical fibers and fiber bragg gratings for damage detection in advanced composite structures: A review, Appl. Mech. Rev. 56 (5) (2003) 493–513, http://dx.doi.org/10.1115/1.1582883.

[16] T. Kori, A. Kori, A. Kori, S. Nandi, A study on fiber bragg gratings and its recent applications, in: Innovative Mobile and Internet Services in Ubiquitous Computing: Proceedings of the 13th International Conference on Innovative Mobile and Internet Services in Ubiquitous Computing, IMIS-2019, Springer, 2020, pp. 880–889.

[17] D. Paschotta, Optical fiber technology- physical principles and applications of different types of optical fibers, Opt. Photonik 52 (2008).

[18] F. Heilmeier, R. Koos, P. Hornberger, J. Hiller, K. Weraneck, M. Jakobi, A.W. Koch, W. Volk, Calibration of cast-in fibre bragg gratings for internal strain measurements in cast aluminium by using neutron diffraction, Meas.: J. Int. Meas. Confed. 163 (2020) http://dx.doi.org/10.1016/j.measurement.2020.107939.

[19] A.V. Kir'yanov, Y.O. Barmenkov, I.L. Martinez, A.S. Kurkov, E.M. Dianov, Opt. Express 14 (2006) http://dx.doi.org/10.1364/oe.14.003981.

[20] T. Erdogan, J.E. Sipe, Tilted fiber phase gratings, J. Opt. Soc. Amer. A 13 (1996) http://dx.doi.org/10.1364/josaa.13.000296.

[21] A. Siekiera, R. Engelbrecht, L. Buethe, B. Schmauss, 1.1-Numerical and experimental investigation of a fiber-optic sensor consisting of a fiber bragg grating in a two-mode fiber for simultaneous sensing of temperature and strain, Proc. OPTO 2013 (2013) 11–16.

[22] R. Paschotta, et al., Encyclopedia of Laser Physics and Technology, vol. 1, Wiley Online Library, 2008.

[23] A. Siekiera, Faser-Bragg-Gitter Und Modeninterferenz in Mehrmodenfasern FÜr Sensoren Zur Simultanen Messung Von Temperatur Und Dehnung, Optische Hochfrequenztechnik Und Photonik, Shaker Verlag, Aachen, 2016.

[24] E.-G. Neumann, Single-Mode Fibers: Fundamentals, vol. 57, Springer, 2013.

[25] J.R. Davis, Tensile Testing, ASM international, 2004.

[26] S. Kammerloher, J. Hoyer, P. Lechner, T. Mittler, W. Volk, A method for characterising the influence of casting parameters on the metallurgical bonding of copper and steel bimetals, Materials 14 (20) (2021) http://dx.doi.org/10.3390/ma14206223, URL https://www.mdpi.com/1996-1944/14/20/6223.

[27] Akhyar, Hot tearing, parameters, and mould types for observation–review, Arch. Foundry Eng. 22 (2) (2022) 25–49.

[28] R. McGill, J.W. Tukey, W.A. Larsen, Variations of boxplots, Amer. Statist. 32 (1) (1978) 12–16.

[29] E. Langford, Quartiles in elementary statistics, J. Stat. Educ. 14 (3) (2006) Retrieved from https://www.tandfonline.com/doi/abs/10.1080/10691898.2006.11910587.

[30] R. Lerch, Messfehler, in: Elektrische Messtechnik: Analoge, Digitale Und ComputergestÜtzte Verfahren, Springer Berlin Heidelberg, Berlin, Heidelberg, 2016, http://dx.doi.org/10.1007/978-3-662-46941-5_5, (S. 103–124).

[31] M.G. Kendall, The Advanced Theory of Statistics (2. Aufl. Vii 521 S.), Charles Griffin and Co. Ltd. London, 1946.

[32] R.A. Fisher, Statistical methods for research workers, in: S. Kotz & N.L. Johnson (Hrsg.), Breakthroughs in Statistics: Methodology and Distribution (S. 66–70), Springer New York, New York, NY, 1992, http://dx.doi.org/10.1007/978-1-4612-4380-9_6.

[33] J. Campbell, The pre-existing microcrack population in metals a personal overview, Procedia Struct. Integr. 43 (2023) 234–239, http://dx.doi.org/10.1016/j.prostr.2022.12.264, Materials structure and micromechanics of fracture. URL https://www.sciencedirect.com/science/article/pii/S2452321622008277.

[34] J. Campbell, Stop pouring, start casting, Int. J. Met. 6 (2012) 7–18.